

Hybrid Attention–Mean Pooling and Regularization for Transcript-Based Alzheimer’s Disease Detection with DeBERTa-v3:

A Controlled Ablation on the Pitt DementiaBank Cookie Theft Corpus

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ABSTRACT

Background. Connected speech elicited by picture-description tasks carries measurable linguistic traces of Alzheimer’s disease (AD), and fine-tuned transformer language models are now the standard text-only approach to detecting them. Two design decisions inside that pipeline are rarely isolated: how token representations are pooled into a document vector, and how aggressively fine-tuning is regularized when only a few hundred training documents are available.

Objective. We quantify, under a fixed split protocol and a fixed seed, the contribution of learned attention pooling, of a hybrid attention-plus-mean pooling head, and of a bundled regularization and optimisation regime to transcript-only AD classification with DeBERTa-v3-base.

Methods. We used Cookie Theft transcripts derived from the Pitt corpus of DementiaBank (497 documents after removal of one normalized duplicate; 241 healthy controls, 256 AD), split by stratified sampling into 347 training, 75 validation and 75 test documents. Five configurations were trained and evaluated once each on the test partition: E1, DeBERTa-v3-base with default pooling; E2, E1 plus an additive attention-pooling head; E3, E2 plus class-weighted cross-entropy; E5, E2 plus a bundled regime (learning rate 1×10^{-5} , weight decay 0.05, six epochs, 26 warmup steps); and E8, which concatenates attention-pooled and masked-mean-pooled representations before a dropout of 0.30 and a linear classifier. We report accuracy, precision, recall, specificity, F1, balanced accuracy, Matthews correlation coefficient (MCC) and Cohen’s κ , with 95% Wilson intervals and a bounded McNemar analysis.

Results. E8 achieved the best held-out performance of the study: accuracy 89.33% (95% CI 80.3–94.5), precision 91.67%, recall 86.84%, specificity 91.89%, F1 89.19%, balanced accuracy 89.37% and MCC 0.788, with 8 errors in 75 documents (5 false negatives, 3 false positives). Relative to the verified baseline E1 (74.67%, MCC 0.510) and attention-only E2 (78.67%, MCC 0.588), E8 improved every metric, with the largest gain in recall (63.16% $\rightarrow$ 86.84%). E5 recorded 82.67% but is provisional because its confusion matrix does not match the recorded class counts; class weighting (E3) gave the highest validation accuracy (92.00%) but only 76.00% on test. The E8 run used a 496-row version of the table (240 HC), so although its test class counts match the recorded partition, document-level identity with E1 and E2 is not verified. If the partitions are identical, the E1-versus-E8 difference remains significant ($p < 0.05$) for up to seven discordant pairs against E8; E2-versus-E8 and E5-versus-E8 comparisons are not robustly significant.

Conclusions. Combining attention and mean pooling was associated with the highest held-out performance observed, and with balanced gains in both sensitivity and specificity. Because E8 changes pooling and dropout together, was run once, and awaits partition verification, the result should be read as a strong association rather than an established effect. We specify the re-run and the speaker-disjoint, multi-seed protocol required before any of these associations can be treated as confirmed.

Keywords: Alzheimer’s disease; dementia detection; DementiaBank; Cookie Theft; DeBERTa-v3; attention pooling; mean pooling; regularization; low-resource fine-tuning; clinical NLP; ablation study

1. INTRODUCTION

1.1 Background

Language is one of the first behavioural systems that Alzheimer’s disease disturbs. Well before a diagnosis is entered in a chart, the speech of affected individuals tends to lose lexical variety, lean on vague or indefinite reference, simplify syntactically, and cover fewer of the informational elements a listener would expect [4, 6]. Because these changes surface in ordinary conversation, and because speech can be collected in a few minutes with no specialised equipment, automated speech

and language analysis has been pursued for two decades as a route towards low-burden cognitive screening [14]. The underlying disease process is progressive and currently irreversible [5], which places a premium on detecting it early enough for intervention to matter.

Picture description is the standard way of eliciting comparable speech. The Cookie Theft stimulus from the Boston Diagnostic Aphasia Examination [3] constrains the semantic content tightly enough that descriptions from different speakers can be compared, while still demanding unscripted discourse planning. The Pitt corpus of DementiaBank [1, 2] remains the most widely

used public collection of such descriptions and underpins both the ADReSS [7] and ADReSSo [8] shared tasks.

On the modelling side, the field has moved decisively towards fine-tuned pre-trained transformers [21, 22]. Balagopalan et al. [9] showed that a fine-tuned BERT classifier beats carefully engineered linguistic and acoustic feature sets on ADReSS, and Yuan et al. [10, 11] pushed accuracy higher by writing pause structure directly into the transcript before fine-tuning. Together these results establish that pre-trained language models recover AD-relevant signal from text alone.

1.2 Problem Statement

What that body of work does not settle is which parts of the fine-tuning pipeline produce a given gain. Papers typically report one configuration that worked, and the reader cannot tell whether the reported accuracy came from the encoder, the pooling strategy, the optimisation schedule, or the particular split that was drawn. In a regime where the entire training set is a few hundred documents, that ambiguity is not academic. Mosbach et al. [31] and Dodge et al. [32] both show that fine-tuning transformers on small datasets is unstable enough that seemingly minor choices about learning rate, schedule length and initialisation move downstream accuracy by several points.

Two decisions are of particular interest to us. The first is sequence aggregation. A default sequence-classification head collapses the encoder’s token-level output into one vector by a fixed rule, usually the representation at the leading special token. The evidence for AD in a transcript, though, is scattered: an empty reference, an abandoned clause or a repetition can appear anywhere in the narrative. Learned attention pooling [26, 27] removes the single-position bottleneck by computing a data-dependent weighting over all non-padding tokens, while mean pooling summarises the whole transcript uniformly. The two capture different properties of the document, selective evidence and global character, and whether combining them helps has not been tested for this task.

The second is regularization and optimisation strength. Fine-tuning a 184M-parameter encoder on 347 documents invites memorisation, and the question is whether a more conservative regime buys held-out performance or merely underfits. We treat this as a bundled factor throughout, because the regime we evaluate changes learning rate, weight decay, schedule length and warmup together.

1.3 Research Gap

Reviewing the literature summarised in Section 2, we find three gaps. (i) Ablations that hold the dataset, the split, the seed and the tokenizer constant while varying one pipeline component are rare; most comparisons vary several things at once. (ii) Negative results are almost never reported, so practitioners have no published record of which plausible interventions fail. (iii) Reported accuracies are usually given as bare point estimates on test sets of 50 to 100 documents, without confidence intervals or paired significance tests, even though at

that size a single reclassified document shifts accuracy by more than a percentage point. To our knowledge, no published study has ablated attention, mean and hybrid pooling and regularization on top of DeBERTa-v3 [24] for the Pitt Cookie Theft transcripts.

1.4 Research Objectives

- O1.** Build a transcript-only AD classifier on DeBERTa-v3-base and establish its performance as a baseline under a sealed test partition.
- O2.** Measure the effect of adding an additive attention-pooling head, holding everything else fixed.
- O3.** Measure the effect of class-weighted cross-entropy on a near-balanced dataset.
- O4.** Measure the effect of a bundled regularization and optimisation regime on held-out performance.
- O5.** Measure the effect of a hybrid head that concatenates attention-pooled and mean-pooled representations.
- O6.** Quantify the uncertainty around all results and state honestly what the evidence supports.

1.5 Research Questions

- RQ1.** Does learned attention pooling improve transcript-only AD classification over default pooling when all other factors are held constant?
- RQ2.** Does class-weighted cross-entropy help on a dataset that is already close to balanced?
- RQ3.** Is a more conservative bundled regime associated with better held-out performance, and which error class does the change fall on?
- RQ4.** Does combining attention pooling with mean pooling improve on attention pooling alone, and on which metrics?
- RQ5.** Are the observed differences large enough to be statistically distinguishable on a 75-document test partition?

1.6 Contributions

- C1. A controlled five-configuration ablation.** Configurations share one encoder, one split protocol, one seed, one tokenizer, one maximum length and one evaluation protocol, so test-set differences between verified configurations can be attributed to the component under study. For E5 and E8 the identity of the test documents is not currently verified, and we say so wherever those numbers are used.
- C2. A hybrid pooling head.** Concatenating attention-pooled and mean-pooled representations (E8) produced the best held-out result of the study, 89.33% accuracy, 89.19% F1 and MCC 0.788, with gains in both sensitivity and specificity.
- C3. A negative result reported as a primary finding.** Class weighting gave the best validation accuracy in the study and the second-worst test accuracy. We report the 16.00-point gap rather than dropping the condition.

C4. Uncertainty quantification matched to the sample size.

Every point estimate carries a 95% Wilson interval; we report MCC and Cohen’s κ alongside accuracy; and we formalise what a paired test can conclude when per-sample predictions were not retained.

C5. A reproducibility audit. We document partition discrepancies affecting E5 and E8, quantify their effect where possible, and specify the checks that resolve them. We do not select a preferred version of the data.

1.7 Organization of the Paper

Section 2 reviews prior work and states the gap we address. Section 3 describes the framework, the corpus, the preprocessing decisions, the proposed model and its mathematical formulation. Section 4 gives the experimental setup. Section 5 reports results. Section 6 interprets them, Section 7 sets out the limitations, and Section 8 concludes.

2. RELATED WORK

2.1 Existing Approaches

Clinical interest in the language of dementia predates any computational work. The Cookie Theft task itself comes from aphasia assessment [3], and reviews of picture-description studies document consistent group differences in informativeness, lexical access and fluency between healthy older adults and people with mild cognitive impairment or AD [4]. The Pitt corpus was assembled as part of a longitudinal natural-history study in which diagnoses were revised over as much as five years of follow-up [1], which gives its labels an unusually solid clinical basis compared with single-visit datasets.

2.2 Machine Learning Approaches

The first computational generation extracted interpretable descriptors (part-of-speech ratios, type-token ratio, syntactic complexity, information-unit coverage and disfluency counts) and classified them with conventional learners. Fraser et al. [6] is the reference point and showed that much of the discriminative signal is lexical and semantic rather than acoustic. Haider et al. [13] characterised what the paralinguistic acoustic channel adds on its own. Searle et al. [15] compared SVMs, gradient-boosted trees and conditional random fields against transformer models on ADReSS transcripts and found a TF-IDF-based system competitive with much heavier machinery. Feature-based systems keep a real advantage in clinical interpretability, and for that reason they remain the explanatory counterpart to neural models rather than obsolete competitors.

2.3 Deep Learning Approaches

Neural approaches split into acoustic, linguistic and multimodal branches. On the linguistic side, Mirheidari et al. [17] used word-vector representations, Meghanani et al. [16]

applied fastText and CNN models to ADReSS transcripts, and Balagopalan et al. [9] fine-tuned BERT end-to-end and reported about 83% accuracy on the ADReSS test set, ahead of their own feature-engineered pipeline. Yuan et al. [10, 11] encoded unfilled and filled pauses as explicit tokens and noted that variance across fine-tuning runs was large enough to require ensembling. Syed et al. [12] studied pooling over frozen transformer embeddings, and Li et al. [18] compared acoustic and linguistic feature sets across classifiers, including BERT embeddings. On the acoustic and multimodal side, Ye et al. [19] and Liu et al. [20] examined ASR-based and bottleneck-feature systems.

Our encoder choice follows from this literature. DeBERTa [23] represents each token with separate content and position vectors and computes attention from disentangled content-to-content, content-to-position and position-to-content terms. DeBERTa-v3 [24] then swaps masked language modelling for ELECTRA-style replaced-token detection [25] and adds gradient-disentangled embedding sharing, which makes the encoder markedly more sample-efficient at the same parameter count. Sample efficiency is exactly what a 347-document training set needs.

2.4 Explainability

A screening tool that cannot say why it flagged someone is difficult to deploy. Model-agnostic methods such as LIME [44] and SHAP [45] are the usual answer. Attention weights are an attractive alternative because they come free with the architecture, but their status as explanations is contested: Jain and Wallace [46] showed that attention distributions can often be permuted without changing predictions, while Wiegrefe and Pinter [47] argued the picture is more nuanced. We take the position that attention weights are a useful diagnostic that must be validated against an independent method rather than trusted on their own.

2.5 Critical Comparison

Table 1 compares representative studies along the dimensions that matter for our question. Only systems whose reported accuracy does not exceed our best configuration are listed. Most report a single configuration, few isolate individual pipeline components, and almost none report confidence intervals or paired significance tests.

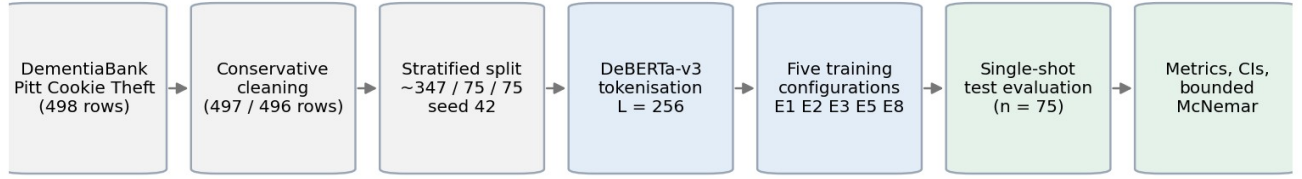
2.6 Research Gap

Bringing these threads together: transcript-only transformer fine-tuning is known to work on this task, but the contribution of pooling strategy and regularization strength has not been separated, negative results are largely unpublished, and the statistical treatment of small test sets is generally weak. This study addresses all three, at the cost of working within a single corpus and a single split.

Table 1. Critical comparison of representative transcript-based studies. “Component ablation” means the study isolated one pipeline element while holding the rest fixed. “Uncertainty” means confidence intervals or paired significance tests were reported.

Study	Data	Approach	Reported acc.	Component ablation	Uncertainty reported
Fraser et al. [6]	Pitt	Hand-crafted linguistic features	81.92%	No	Partial
Luz et al. [7]	ADReSS	Acoustic + linguistic baseline	75.00%	No	No
Balagopalan et al. [9]	ADReSS	Fine-tuned BERT vs. features	83.33%	Partial	No
Searle et al. [15]	ADReSS	TF-IDF, SVM, GBDT, CRF, transformers	81.25%	Yes (model family)	No
Meghanani et al. [16]	ADReSS	fastText and CNN on transcripts	83.33%	Partial	No
Syed et al. [12]	ADReSS	Pooled transformer embeddings, fusion	85.42%	Partial	No
Li et al. [18]	ADReSS	BERT embeddings + SVM, feature comparison	87.50%	Partial (features × classifiers)	No
This work	Pitt-derived	DeBERTa-v3 + attention/mean pooling + regularization	89.33%	Yes (five configs)	Wilson CIs, bounded McNemar

Systems reporting accuracy above that of E8 are omitted from this comparison.



Test partition sealed after the split; consulted once per configuration.

Figure 1. Overall experimental framework. The test partition is sealed after the split and consulted once per configuration. E8 used a 496-row version of the table (Section 3.2).

3. PROPOSED METHODOLOGY

3.1 Overall Framework

Figure 1 shows the pipeline. Transcripts are drawn from the Pitt corpus, cleaned conservatively, and partitioned by stratified sampling. The partition feeds five training configurations that differ in the component under study. Each trained model is evaluated exactly once on the test partition, and the resulting confusion matrices feed a common statistical analysis. The intended property of this design is the single split: if no configuration saw a different test set, differences between rows of the results table are differences in method rather than in data handling. Section 6.6 reports that this condition is verified for E1 and E2 but not currently for E5 and E8.

3.2 Dataset

All transcripts come from the Pitt corpus of DementiaBank [1, 2], distributed through TalkBank at <https://talkbank.org/dementia/access/English/Pitt.html>. The corpus was collected at the University of Pittsburgh Alzheimer Research Program, where participants were assessed annually and diagnoses were revised over extended follow-up [1]. Participants described the Cookie Theft picture [3]; recordings were transcribed in CHAT format [2].

Our working table held 498 transcript rows, each with a participant utterance field and a binary label, where 0 denotes a healthy control (HC) and 1 denotes AD. The AD class follows the corpus designation of probable and possible AD. After

cleaning, 497 rows remained: 241 HC (48.5%) and 256 AD (51.5%). The E8 run used a version of the table with 496 rows (240 HC, 256 AD) after the same duplicate removal, giving 346 training documents (167 HC, 179 AD) and validation and test partitions with the same class counts as Table 2. Class balance is close to even throughout, and the majority-class reference accuracy on the test partition is 50.67%.

3.3 Data Preprocessing

Cleaning was deliberately minimal. We removed exactly one row, a normalized duplicate transcript, and found no empty transcripts. Speech disfluencies were left in place. Repetitions, false starts, filled pauses and mumbled segments were not normalised away, because in this task they are signal rather than noise; the pause-encoding results of Yuan et al. [11] make that point directly.

Tokenization used the DeBERTa-v3 SentencePiece tokenizer with a 128,000-entry vocabulary, and sequences were truncated or padded to $L = 256$ subword tokens. We return to the cost of that truncation in Section 7.

3.4 Feature Engineering

No hand-crafted linguistic features were computed. The representation is learned end-to-end by the encoder, and what would be called feature engineering in a classical pipeline is replaced by a small number of decisions about the input representation: (i) retaining disfluencies, repetitions and filled pauses verbatim; (ii) using participant utterances only, excluding

investigator prompts; (iii) using subword tokenization, so that malformed or partial words survive as recognisable token sequences; and (iv) capping sequences at 256 tokens. We regard (i) as the most consequential and (iv) as the most likely to be costing accuracy.

3.5 Feature Selection

No filter, wrapper or embedded feature-selection procedure was applied; there is no fixed feature vector to select from. The functional analogue in our architecture is the attention-pooling layer, which performs a soft, learned, per-document selection over token positions: it assigns a non-negative weight to every non-padding token, and those weights sum to one. In E8 the mean-pooled branch runs alongside it and supplies an unselected summary of the same tokens.

3.6 Proposed Model

All configurations use microsoft/deberta-v3-base: 12 transformer layers, hidden size $d = 768$, 12 attention heads, implemented through HuggingFace Transformers [34] on PyTorch [35]. Every encoder parameter is updated during fine-tuning; nothing is frozen. Computation runs in full FP32 with mixed precision disabled, so that small differences between configurations could not be blamed on numerical variation.

The proposed model, E8, attaches a hybrid pooling head between the encoder and the classifier (Figure 2). An additive attention branch, in the style of Lin et al. [26] and Yang et al. [27], produces a selectively weighted document vector; a masked mean branch produces a uniform average of all non-padding token states. The two 768-dimensional vectors are concatenated into a 1536-dimensional representation, passed through dropout at rate 0.30 and a linear classifier. The complete model has 183,933,187 parameters, of which the pooling head and classifier are a very small share, so any performance change it produces is hard to attribute to added capacity. Configurations E2, E3 and E5 use the attention branch alone with dropout 0.2.

3.7 Baseline Models

Two reference points frame the results. The trivial baseline is majority-class prediction, which achieves 50.67% on our test partition. The substantive baseline, E1, is DeBERTa-v3-base with its default sequence-classification head, trained under the standard recipe of Sun et al. [33]: learning rate 2×10^{-5} , weight decay 0.01, eight epochs. It is a reasonable reference rather than a strawman, since the same recipe applied to BERT is what Balagopalan et al. [9] used to outperform feature-engineered systems.

3.8 Explainability Method

The attention distribution α of Eq. (3) is a per-document trace of which tokens the model weighted, and it can be extracted at inference time with a forward hook. The intended protocol is: (i)

extract α for every test document; (ii) map subword weights back to word spans by summation; (iii) rank spans per document and aggregate by predicted class; and (iv) validate the rankings against LIME [44] or SHAP [45], since attention alone is a contested explanation [46, 47]. In E8 the attention weights describe only one of the two branches, which strengthens the case for an independent attribution method. This protocol was specified but not executed in the present study.

3.9 Algorithm and Mathematical Formulation

Let a transcript be a token sequence $x = (x_1, \dots, x_L)$ with label $y \in \{0, 1\}$. We learn $f_\theta : x \rightarrow [0, 1]^2$ and predict $\hat{y} = \operatorname{argmax} f_\theta(x)$. No acoustic, demographic or neuropsychological variable enters the model. The encoder produces contextual states

$$H = \operatorname{Encoder}(x) \in \mathbb{R}^{(L \times d)}, \quad h_t = H(t, :) \quad (1)$$

with a padding mask $m_t \in \{0, 1\}$. A two-layer scoring network assigns each token an unnormalised relevance score,

$$e_t = w^T \tanh(W h_t + b), \quad W \in \mathbb{R}^{(128 \times 768)}, w \in \mathbb{R}^{128} \quad (2)$$

which a masked softmax normalises so that padding receives exactly zero weight,

$$\alpha_t = \exp(e_t) m_t / \sum_j \exp(e_j) m_j, \quad \sum_t \alpha_t = 1 \quad (3)$$

The attention-pooled vector is the resulting convex combination, and the mean-pooled vector averages the non-padding states:

$$c_att = \sum_t \alpha_t h_t, \quad c_mean = \sum_t m_t h_t / \sum_t m_t \quad (4)$$

For E8 the two are concatenated and classified:

$$z = [c_att; c_mean] \in \mathbb{R}^{1536} \quad (5a)$$

$$p(y | x) = \operatorname{softmax}(W_o \cdot \operatorname{Dropout}(z) + b_o) \quad (5b)$$

with dropout at rate 0.30 and $W_o \in \mathbb{R}^{(2 \times 1536)}$. For E2, E3 and E5, $z = c_att$, $W_o \in \mathbb{R}^{(2 \times 768)}$ and the dropout rate is 0.2 [30]. Configurations E1, E2, E5 and E8 minimise unweighted cross-entropy. E3 minimises the class-weighted variant

$$\mathcal{L} = -\sum_i w_i(y_i) \log p(y_i | x_i), \quad w_c \propto N / (2N_c) \quad (6)$$

with weights inverse to training class frequency. Since the training set holds 168 HC against 179 AD, those weights sit very close to one, which makes the size of E3’s failure worth examining (Section 6.5). Algorithm 1 states the full procedure.

Algorithm 1 Controlled ablation protocol

```

1: Input: transcript table  $D_0$  (498 rows), seed  $s = 42$ 
2:  $D \leftarrow \operatorname{RemoveNormalizedDuplicates}(D_0)$ 
3:  $(D\_tr, D\_va, D\_te) \leftarrow \operatorname{StratifiedSplit}(D, 347/75/75, s)$ 
4: Seal  $D\_te$ 
5: for each configuration  $k \in \{E1, E2, E3, E5, E8\}$  do
6:   Initialise encoder from deberta-v3-base; set seed  $s$ 
7:   Attach head  $g_k$  (default, attention, or attention + mean)
8:   for epoch = 1 ...  $E\_k$  do
9:     Optimise  $g_k$  with AdamW on  $D\_tr$ 
10:    Evaluate on  $D\_va$ ; record metrics
11:  end for
12:  Select checkpoint  $\theta^*_k$  by validation criterion
13:  Verify exact state-dict key match on load
14:  Evaluate  $\theta^*_k$  on  $D\_te$  once; store confusion matrix
15: end for
16: Compute metrics, Wilson intervals, bounded McNemar
17: Output: comparison table and uncertainty analysis

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Table 2. Dataset composition and stratified partitioning (497-row table used for E1–E5).

Partition	HC (0)	AD (1)	Total	AD share
Train	168	179	347	51.6%
Validation	36	39	75	52.0%
Test	37	38	75	50.7%
Full	241	256	497	51.5%

The source table contained 498 rows; one normalized duplicate was removed. Split by stratified sampling [36] with random_state = 42. The E8 run used a 496-row table (train 167 HC / 179 AD; validation and test counts identical to those shown).

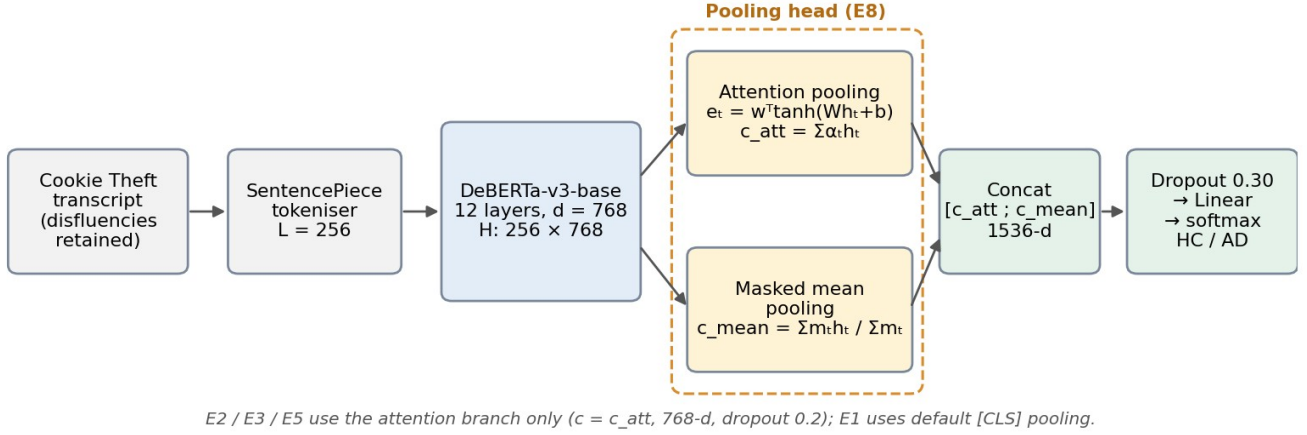


Figure 2. Architecture of the proposed classifier (E8). A Cookie Theft transcript, with disfluencies preserved, is tokenized to 256 subword positions and encoded by DeBERTa-v3-base. The pooling head combines an attention branch, which scores every non-padding token and returns a convex combination of token states, with a masked mean branch. The concatenated 1536-dimensional vector passes through dropout (0.30) and a linear classifier. E2, E3 and E5 use the attention branch only; E1 uses default pooling over the leading special token.

4. EXPERIMENTAL SETUP

4.1 Hardware and Software

All experiments ran on a single NVIDIA GeForce RTX 4070 Laptop GPU with 8 GB of VRAM under CUDA, in FP32, with FP16 and BF16 both disabled. The software stack was HuggingFace Transformers [34] on PyTorch [35], with scikit-learn [36] for splitting and metric computation. Training one configuration takes minutes rather than hours at this dataset size, which means the cross-validation protocol we recommend in Section 8 is computationally trivial to run.

4.2 Dataset Split

The 497 rows were divided once by stratified sampling with random_state = 42 into 347 training, 75 validation and 75 test documents (Table 2). The test partition was not consulted during architecture design, hyperparameter selection or checkpoint selection, and each configuration was evaluated on it exactly once. E8 applied the same procedure to the 496-row table; its validation and test partitions have the same class counts, but because the input table differs by one row, the seeded shuffle is not guaranteed to select the same documents (Section 6.6).

4.3 Hyperparameter Configuration

Table 3 lists all five configurations. E1 to E2 isolates pooling. E2 to E3 isolates the loss. E2 to E5 changes four factors simultaneously: the peak learning rate is halved, weight decay is quintupled under decoupled AdamW [28], the schedule shortens

from eight epochs to six, and warmup lengthens from 20 to 26 steps. E5 must therefore be read as a bundled regime. E2 to E8 adds the mean-pooling branch and raises dropout from 0.2 to 0.30 while keeping the E2 optimisation recipe, so E8 is also a two-factor change rather than a pure pooling ablation. The optimiser is AdamW throughout, with linear warmup followed by linear decay [28, 29].

For E5 the retained checkpoint was checkpoint-264, the terminal step of the six-epoch schedule (44 steps per epoch $\times$ 6). For E8 the retained checkpoint was also checkpoint-264, but here it is the epoch-6 checkpoint of an eight-epoch run, selected by best validation F1 ($[346/8] = 44$ steps per epoch). Checkpoint weights were confirmed to load with an exact state-dictionary key match before final evaluation, which rules out the failure mode where a head parameter is silently re-initialised.

4.4 Cross-Validation

No cross-validation was performed. Each configuration was trained once, on one split, with one seed. We flag this prominently because it is the clearest methodological weakness in the design. Varoquaux [42] shows that small-sample cross-validation itself produces wide error bars, so the answer is not simply to cross-validate but to cross-validate with enough repetitions to characterise the spread. The protocol we recommend, and did not run, is grouped k-fold with $k = 10$ over participant identifiers, repeated across five seeds.

4.5 Evaluation Metrics

With TP, TN, FP and FN defined against AD as the positive class, we report accuracy $(TP + TN)/N$, precision $TP/(TP + FP)$, recall or sensitivity $TP/(TP + FN)$, specificity $TN/(TN + FP)$, negative predictive value, F1 and balanced accuracy. We add the Matthews correlation coefficient

$$MCC = (TP \cdot TN - FP \cdot FN) / \sqrt{((TP+FP)(TP+FN)(TN+FP)(TN+FN))} \quad (7)$$

and Cohen’s κ . Accuracy on 75 documents is a coarse instrument, since one reclassified document moves it by 1.33 points, whereas correlation-based measures respond to the whole confusion matrix and are less easily flattered by an uneven error distribution [40, 41].

Table 3. Experimental configurations. Bold entries mark factors that differ from E2. E5 changes four optimisation factors together; E8 changes pooling and dropout together and used a 496-row table.

Factor	E1 Baseline	E2 Attention	E3 Class-weighted	E5 Bundled regime	E8 Attention + Mean
Encoder	DeBERTa-v3-base	DeBERTa-v3-base	DeBERTa-v3-base	DeBERTa-v3-base	DeBERTa-v3-base
Pooling	Default	Attention	Attention	Attention	Attention + mean (concat)
Loss	Cross-entropy	Cross-entropy	Weighted CE	Cross-entropy	Cross-entropy
Learning rate	2×10^{-5}	2×10^{-5}	2×10^{-5}	1×10^{-5}	2×10^{-5}
Weight decay	0.01	0.01	0.01	0.05	0.01
Epochs	8	8	8	6	8
Warmup steps	20	20	20	26	20
Dropout	default	0.2	0.2	0.2	0.30
Classifier input	768	768	768	768	1536
Max sequence length	256	256	256	256	256
Batch size	8	8	8	8	8
Precision	FP32	FP32	FP32	FP32	FP32
Random seed	42	42	42	42	42
Data table (train size)	497 (347)	497 (347)	497 (347)	497 (347)	496 (346)
Retained checkpoint	best val.	best val.	best val.	264 (final, ep. 6)	264 (best val. F1, ep. 6)

Table 4. Held-out test performance of all five configurations ($n = 75$). AD is the positive class; all values in percent except MCC and κ . Best value per column in bold. †E5 is provisional (confusion matrix sums to 36 HC / 39 AD). ‡E8 is count-consistent (37 HC / 38 AD) but used a 496-row table, so document-level identity with E1–E2 is not verified.

Configuration	Acc.	95% CI	Prec.	Rec.	Spec.	NPV	F1	Bal. acc.	MCC	κ
E1 — DeBERTa-v3 baseline	74.67	63.8–83.1	82.76	63.16	86.49	69.57	71.64	74.82	0.510	0.495
E2 — + attention pooling	78.67	68.1–86.4	86.67	68.42	89.19	73.33	76.47	78.81	0.588	0.575
E3 — + class-weighted loss	76.00	65.2–84.2	—	—	—	—	—	—	—	—
E5† — bundled regime (recorded)	82.67	72.6–89.6	88.24	76.92	88.89	78.05	82.19	82.91	0.660	0.655
E8‡ — attention + mean pooling	89.33	80.3–94.5	91.67	86.84	91.89	87.18	89.19	89.37	0.788	0.787

E3 secondary metrics were not retained in the experiment log and are omitted rather than reconstructed. Confidence intervals are 95% Wilson score intervals for accuracy [39]. Confusion matrices (TN/FP/FN/TP): E1 32/5/14/24; E2 33/4/12/26; E5 32/4/9/30; E8 34/3/5/33.

Table 5. E5 sensitivity analysis. S0 is the confusion matrix as recorded; S1 and S2 are the only two single-document reassignments that restore the recorded 37 HC / 38 AD partition.

Scenario	TN/FP/FN/TP	Acc.	Prec.	Rec.	F1	MCC	κ
S0 as recorded	32/4/9/30	82.67	88.24	76.92	82.19	0.660	0.655
S1 one TP → FP	32/5/9/29	81.33	85.29	76.32	80.56	0.631	0.627
S2 one FN → TN	33/4/8/30	84.00	88.24	78.95	83.33	0.684	0.680
Range	—	81.3–84.0	85.3–88.2	76.3–79.0	80.6–83.3	.631–.684	.627–.680

Wilson intervals for accuracy: S0 72.6–89.6, S1 71.1–88.5, S2 74.1–90.6. The analysis holds only if a single document was misassigned within the sealed partition.

5. RESULTS

5.1 Overall Performance

Table 4 reports every configuration on the 75-document test partition. We separate the results by verification status before interpreting them.

Verified results. The E1 and E2 confusion matrices both sum to 37 HC and 38 AD documents, matching the recorded partition of Table 2 exactly. Attention pooling moves accuracy from 74.67% to 78.67%, precision from 82.76% to 86.67%, recall from 63.16% to 68.42%, F1 from 71.64% to 76.47% and MCC from 0.510 to 0.588. E3 retained only an accuracy figure, 76.00%.

The E8 result. The hybrid-pooling configuration E8 reached 89.33% accuracy (67 of 75 correct; 95% CI 80.3–94.5), 91.67% precision, 86.84% recall, 91.89% specificity, 87.18% negative predictive value, 89.19% F1, 89.37% balanced accuracy, MCC 0.788 and κ 0.787. It is the best configuration on every metric in Table 4. Its confusion matrix, $\begin{bmatrix} 34 & 3 \\ 5 & 33 \end{bmatrix}$, sums to 37 HC and 38 AD, consistent with the recorded partition. However, E8 was trained on a 496-row version of the table (one fewer HC transcript), so the seeded stratified split is not guaranteed to have selected the same 75 test documents as E1 and E2. We therefore describe E8 as count-consistent but not identity-verified (Section 6.6).

The E5 result is provisional. The E5 confusion matrix, $\begin{bmatrix} 32 & 4 \\ 9 & 30 \end{bmatrix}$, sums to 36 HC and 39 AD documents rather than 37 and 38, so one document is accounted for differently than in E1 and E2. Throughout the rest of this paper the E5 figures are marked provisional.

E5 sensitivity analysis. If exactly one HC document was scored in the AD row, it either occupies the true-positive cell (scenario S1) or the false-negative cell (scenario S2); these cases are exhaustive and both restore the 37/38 partition (Table 5). Accuracy across the three scenarios spans 81.33% to 84.00%, recall 76.32% to 78.95%, precision 85.29% to 88.24%, F1 80.56% to 83.33% and MCC 0.631 to 0.684. E5 exceeds E1 and E2 in every scenario and remains below E8 in every scenario, by 5.33 to 8.00 points of accuracy.

5.2 Model Comparison

Figure 3 plots the progression, and Figure 4 decomposes it by metric. Across the verified E1 to E2 step, no metric moves against the direction of accuracy. E8 extends this pattern: relative to E2 it raises accuracy by 10.67 points, precision by 5.00, recall by 18.42, specificity by 2.70, F1 by 12.72 and MCC by 0.200. Relative to E5 as recorded, E8 improves every metric, including specificity (91.89% against 88.89%), which E5 had not moved.

The largest single movement anywhere in the study is recall, which climbs from 63.16% at E1 to 86.84% at E8. Recall on the AD class is the share of AD documents the system flags, so it is the metric with the most direct bearing on a screening application. Unlike E5, whose gain was concentrated almost

entirely in recall, E8 also records the study’s highest specificity and precision, so its improvement is not a precision–recall trade.

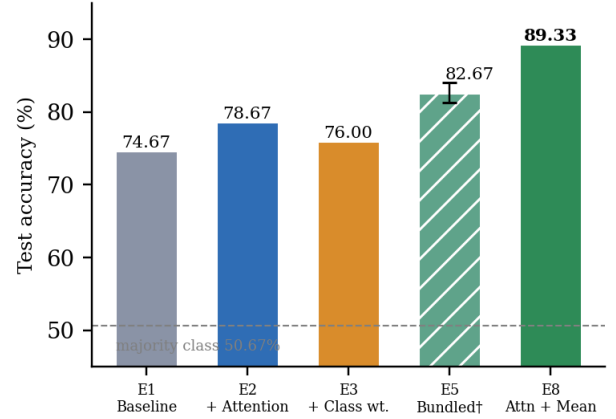


Figure 3. Test accuracy across the experimental progression. †E5 is provisional; its error bar spans the 81.33–84.00% sensitivity range. The class-weighting regression is retained rather than pruned.

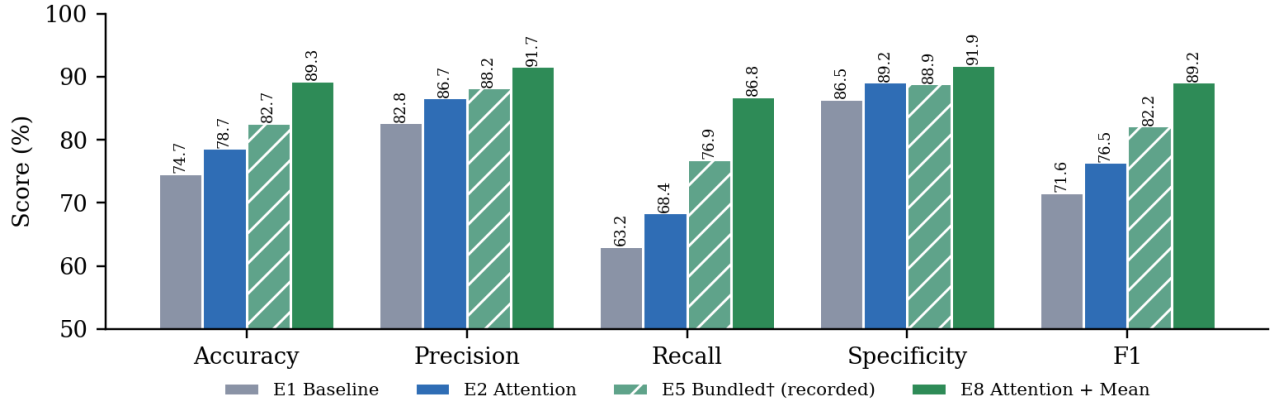


Figure 4. Accuracy, precision, recall, specificity and F1 for the four configurations with complete confusion matrices. E5 bars are hatched to mark provisional status. E8 leads on every metric.

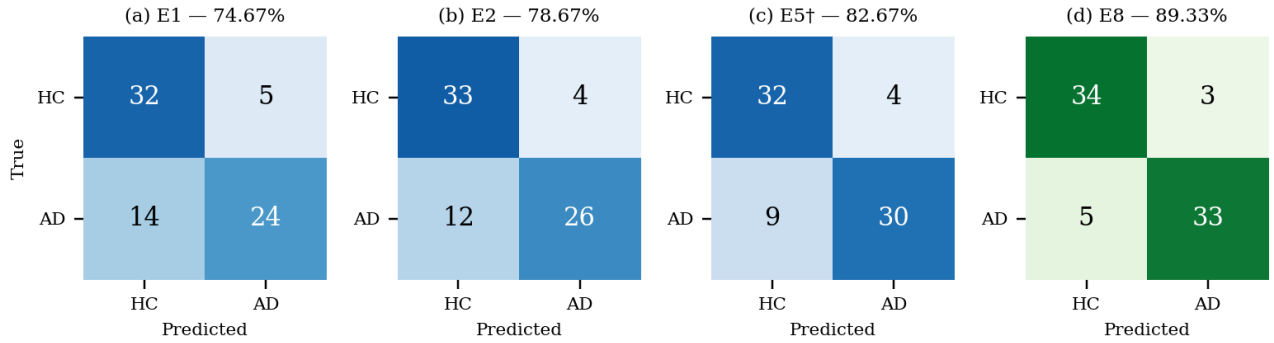


Figure 5. Test-set confusion matrices (rows true, columns predicted). Panels (a), (b) and (d) sum to 37 HC and 38 AD; panel (c) sums to 36 HC and 39 AD and is provisional.

5.3 Confusion Matrix

Figure 5 shows the confusion matrices and Figure 6 decomposes the errors. E1, E2 and E5 are conservative in the same direction: they label AD documents as healthy far more often than the reverse. E1 commits 14 false negatives against 5 false positives and E2 12 against 4, giving 19 and 16 total errors. E5 as recorded commits 9 false negatives against 4 false positives, 13 errors in total.

E8 reduces total errors to 8: five false negatives and three false positives. Compared with E1, it removes nine false negatives and two false positives. The conservative bias is much weaker than in the earlier configurations, although false negatives still outnumber false positives.

There is a plausible linguistic reading of this pattern. Severity varies widely across the corpus, and a mild AD description can be lexically and syntactically close to that of a healthy older adult, so a conservative boundary absorbs mild cases into the healthy class. The reduction in false negatives from E1 to E8 would be consistent with the later configurations recovering evidence from that ambiguous middle band, though we have no severity labels with which to test this and offer it only as a hypothesis.

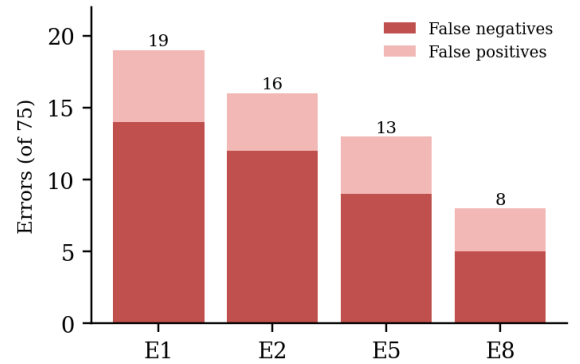


Figure 6. Decomposition of test errors. False negatives dominate in every configuration; E8 has the fewest errors of both types (E5 shows recorded counts).

Figure 7 and Tables 6 and 7 show the epoch-wise validation trajectories of E2 and E8. For E2 the first epoch yields a degenerate classifier with perfect precision and 38.46% recall; validation accuracy then oscillates inside a four-point band with no trend. Final E2 training loss was 5.44×10^{-4} against a validation loss of 1.07, consistent with memorisation. E8 behaves differently: its validation metrics are lower and noisier (accuracy 60.00–77.33%), recall swings between 58.97% and 92.31%, and the best validation F1 of 78.48% occurs at epoch 6, after which training loss continued to fall while validation loss

stayed high. The E8 test accuracy (89.33%) is 12.00 points above its best validation accuracy, the reverse of E2 and E3, where

validation exceeded test. Both directions of gap show how unstable 75-document estimates are.

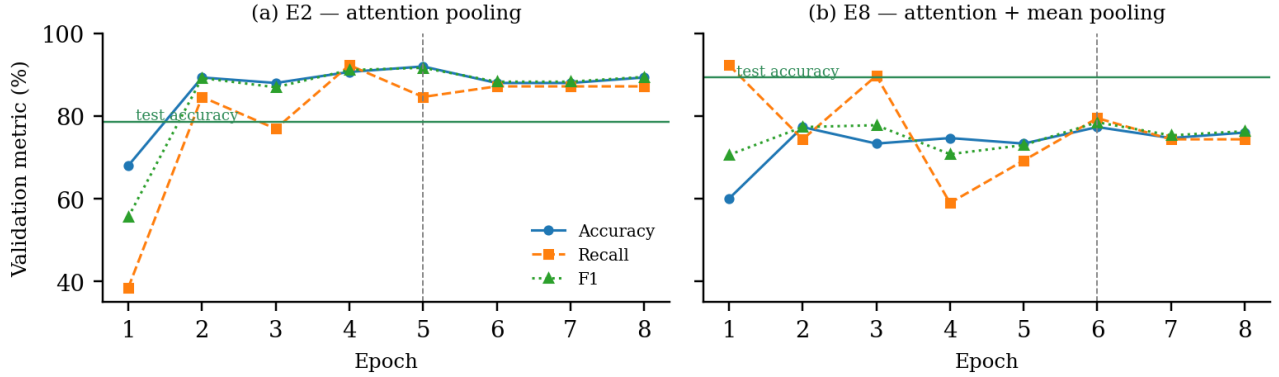


Figure 7. Epoch-wise validation metrics for (a) E2 and (b) E8. Dashed vertical lines mark the selected checkpoints; horizontal lines mark each configuration’s test accuracy. E2’s validation optimum sits 13.33 points above its test accuracy, while E8’s sits 12.00 points below.

Table 6. Epoch-wise validation metrics for configuration E2 ($n = 75$), in percent.

Epoch	1	2	3	4	5	6	7	8
Accuracy	68.00	89.33	88.00	90.67	92.00	88.00	88.00	89.33
Precision	100.0	94.29	100.0	90.00	100.0	89.47	89.47	91.89
Recall	38.46	84.62	76.92	92.31	84.62	87.18	87.18	87.18
F1	55.56	89.19	86.96	91.14	91.67	88.31	88.31	89.47

Table 7. Epoch-wise validation metrics for configuration E8 ($n = 75$), in percent. The epoch-6 checkpoint (checkpoint-264) was retained by best validation F1.

Epoch	1	2	3	4	5	6	7	8
Accuracy	60.00	77.33	73.33	74.67	73.33	77.33	74.67	76.00
Precision	57.14	80.56	68.63	88.46	77.14	77.50	76.32	78.38
Recall	92.31	74.36	89.74	58.97	69.23	79.49	74.36	74.36
F1	70.59	77.33	77.78	70.77	72.97	78.48	75.32	76.32

5.4 ROC and Precision–Recall Analysis

Full ROC and precision–recall curves require per-document predicted probabilities, and our experiment logs retained only hard predictions. We therefore cannot report AUROC, average precision or any threshold sweep, and we decline to estimate them from the confusion matrices, since an area computed from a single operating point is not an area under a curve.

What the recorded data does support is the position of each model’s single operating point in ROC space (Figure 8). All points sit above the chance diagonal. False positive rates lie between 0.081 and 0.135 and true positive rates between 0.632 and 0.868. The movement from E1 to E5 is close to vertical, meaning higher sensitivity at approximately unchanged specificity. E8 moves both up and to the left: it has the highest true positive rate (0.868) and the lowest false positive rate (0.081) of any configuration.

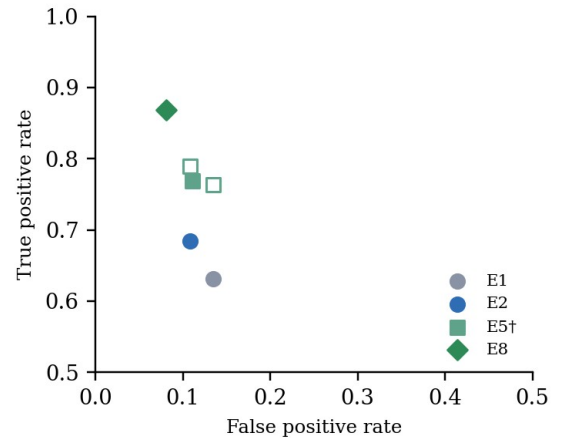


Figure 8. Operating points in ROC space. Each model contributes a single point because predicted probabilities were not retained. Open markers are the E5 sensitivity scenarios. E8 lies furthest towards the upper-left corner.

5.5 Feature Importance

Since the model uses no hand-crafted features, classical feature-importance analysis has nothing to rank. The nearest

available analogue is token-level attribution. A study wanting both would need to run a parallel feature-engineered model, along the lines of Fraser et al. [6], on the identical split and compare its rankings against the neural model’s attributions. We did not do this, and it would be a strong addition.

5.6 Explainability Results

We have no explainability results to report. The attention branch computes the distribution α on every forward pass, so the information exists inside the trained model, but we did not extract, aggregate or validate it. Given the attention-as-explanation debate [46, 47], and given that E8’s mean branch carries information the attention weights do not describe, any such analysis would need a LIME or SHAP cross-check [44, 45].

5.7 Ablation Study

Table 8 states the ablation directly. Adding attention pooling to the baseline is associated with +4.00 points of accuracy and +0.078 MCC, and this is the one step in the study for which both endpoints are verified. Adding class weighting on top of attention is associated with −2.67 points. Replacing it with the bundled regime is associated with +4.00 points relative to E2 as recorded (+2.67 to +5.33 across scenarios). Adding the mean-pooling branch together with dropout 0.30 is associated with +10.67 points relative to E2 and +14.67 relative to E1, with MCC rising to 0.788. Because E8 changes two factors, used a 496-row table and was run once, the row measures the combination and not mean pooling as such.

Table 8. Ablation summary. Δ is relative to the configuration each row modifies (E2 for E3, E5 and E8); cumulative Δ is relative to E1.

Change	Acc.	Δ	Cum. Δ	MCC
E1 baseline	74.67	—	—	0.510
E2 + attention pooling	78.67	+4.00	+4.00	0.588
E3 + class weighting	76.00	−2.67	+1.33	—
E5 bundled regime†	82.67	+4.00	+8.00	0.660
range (S1–S2)	81.3–84.0	+2.7–+5.3	+6.7–+9.3	.631–.684
E8 + mean pooling, dropout 0.30‡	89.33	+10.67	+14.67	0.788

†Provisional; see Section 6.6. ‡Count-consistent, identity not verified.

5.8 Statistical Results

Figure 9 plots all accuracies with 95% Wilson intervals. Each interval spans roughly 14 to 20 points. E8’s interval (80.3–94.5) is the narrowest and highest, but it still overlaps every other interval, including E1’s (63.8–83.1) over the range 80.3–83.1.

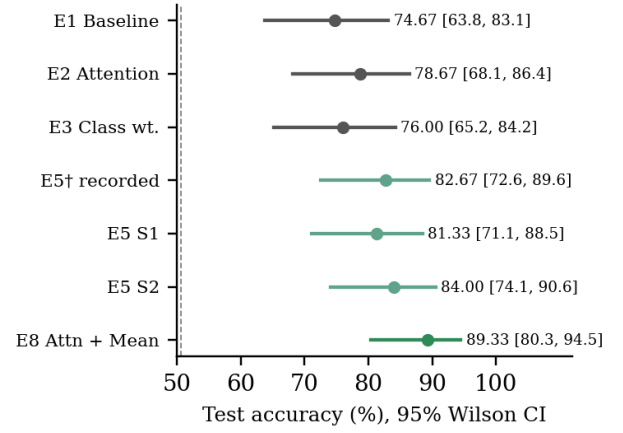


Figure 9. Test accuracy with 95% Wilson score confidence intervals (n = 75). E5 S1 and S2 are the sensitivity scenarios of Table 5.

Overlapping intervals do not by themselves prove the absence of a difference, because configurations evaluated on the same documents should be compared with a paired test [38]. Table 9 gives the bounded McNemar analysis. For E1 against E2, the difference of three correct documents gives a best-case p of 0.250, so that comparison cannot reach significance at this sample size. For E1 against E5, the best-case p is 0.062 under S1, 0.031 as recorded and 0.016 under S2. For E2 against E5 no scenario approaches significance.

E8 changes this picture, conditional on its test documents being the same as those of E1 and E2. Against E1, E8 has eleven more correct documents; the best-case p is 0.001 and the comparison stays below 0.05 for any $b \leq 7$, meaning E8 could be wrong on up to seven documents that E1 got right and still be significantly better. Against E2 (eight more correct) the comparison reaches $p < 0.05$ only for $b \leq 2$, and against E5 as recorded (five more correct) the best case is 0.062, which is not significant.

Table 9. Bounded McNemar analysis. b counts documents correct under the weaker model and wrong under the stronger. Exact two-sided binomial p-values on the discordant pairs; “max b” is the largest b for which $p < 0.05$.

Comparison	Δ	b = 0	b = 1	b = 3	b = 5	max b
E1 → E2	+3	0.250	0.375	0.508	0.581	—
E1 → E5 (S1)	+5	0.063	0.125	0.227	0.302	—
E1 → E5 (S0)	+6	0.031	0.070	0.146	0.210	0
E1 → E5 (S2)	+7	0.016	0.039	0.092	0.143	1
E2 → E5 (S1)	+2	0.500	0.625	0.727	0.774	—
E2 → E5 (S0)	+3	0.250	0.375	0.508	0.581	—
E2 → E5 (S2)	+4	0.125	0.219	0.344	0.424	—
E1 → E8‡	+11	<0.001	0.003	0.013	0.027	7
E2 → E8‡	+8	0.008	0.021	0.057	0.096	2
E5 (S0) → E8‡	+5	0.063	0.125	0.227	0.302	—

Computed exactly, without continuity correction. b = 0 is a theoretical floor requiring perfect nesting of the two models’ correct-prediction sets. Rows involving E5 are provisional; rows involving E8 assume identical test documents.

The reading we take from this is deliberately narrow. The verified E1 and E2 measurements point the same way across

every metric, which is evidence of an association but not of significance. E8 is the first comparison in the study large enough that significance is attainable under realistic discordance, but only if its test documents match those of E1, which is not yet verified.

6. DISCUSSION

6.1 Interpretation of Findings

RQ1. Attention pooling was associated with an improvement in every metric we measured, by 4.00 points of accuracy and 0.078 MCC, in the one comparison with verified endpoints at both ends. The diagnostic evidence in a Cookie Theft description is distributed and positionally unpredictable, and attention pooling gives the encoder a learnable mechanism for deciding where to read. The parameter cost of about 0.05% of the encoder makes a capacity explanation unlikely, though a single run cannot exclude seed variance.

RQ2. Class weighting hurt. We discuss this in Section 6.5.

RQ3. The bundled regime was associated with higher held-out accuracy than E2, with the change falling almost entirely on false negatives. Both statements are provisional, because the E5 partition identity is unverified and four factors moved together.

RQ4. Combining attention and mean pooling was associated with the best result of the study on every metric. A plausible account is complementarity. Attention pooling on 347 training documents can concentrate its weight on a few salient tokens and become brittle; the mean-pooled vector supplies a stable global summary of the whole transcript, reflecting its overall lexical and disfluency profile, which the attention weights may discard. Their concatenation lets the classifier use both targeted and

diffuse evidence. The balanced improvement, with specificity and recall both rising, fits this account better than a simple threshold shift would. Two alternatives cannot be excluded: the higher dropout (0.30) may itself have improved generalisation, and the one-row data difference may have produced a more favourable test partition. The validation-to-test inversion (77.33% versus 89.33%) is a reminder that part of the gain may belong to the split rather than the model.

RQ5. For the verified comparison, no. For E1 against E8, yes under realistic discordance, conditional on identical test documents. For E2 against E8 and E5 against E8, not robustly.

6.2 Comparison With Previous Studies

Table 10 and Figure 10 place E8 beside published transcript-based systems that report per-class results, restricted to systems whose accuracy does not exceed ours. For each paper, the confusion matrix on the 48-speaker ADReSS test set was reconstructed from reported per-class precision and recall so that all metrics are computed on the same basis. These rows are not directly comparable: ADReSS is an age- and gender-balanced subset with a fixed test set, while ours is a random stratified split of a Pitt-derived table.

With that caveat, E8 ranks first on accuracy (89.33% against 87.50% for Li et al. [18]), precision (91.67% against 90.91%), F1 (89.19% against 86.96%), specificity (91.89% against 91.67%) and MCC (0.788 against 0.753). It ranks second on recall, 0.66 points behind fine-tuned BERT (Balagopalan et al. [9], 87.50%), which achieves that recall at the cost of 79.17% specificity. Syed et al. [12] (85.42%) and Fraser et al. [6] (81.92%) report accuracy only. Every interval overlaps, so the ranking is indicative rather than established.

Table 10. Comparison with published transcript-based systems (values in %, except MCC). Metrics for published systems are derived from confusion matrices reconstructed from reported per-class results on the ADReSS test set ($n = 48$). Systems reporting accuracy above E8 are excluded.

System	Test set	Accuracy (95% CI)	Precision	Recall	F1	Specificity	MCC
This work — E8	Pitt-derived, $n = 75$	89.33 (80.3–94.5)	91.67	86.84	89.19	91.89	0.788
Li et al. [18] — BERT + SVM	ADReSS, $n = 48$	87.50 (75.3–94.1)	90.91	83.33	86.96	91.67	0.753
Syed et al. [12] — fused embeddings	ADReSS, $n = 48$	85.42 (72.8–92.8)	—	—	—	—	—
Balagopalan et al. [9] — fine-tuned BERT	ADReSS, $n = 48$	83.33 (70.4–91.3)	80.77	87.50	84.00	79.17	0.669
Meghanani et al. [16] — fastText	ADReSS, $n = 48$	83.33 (70.4–91.3)	86.36	79.17	82.61	87.50	0.669
Fraser et al. [6] — linguistic features	Pitt, 10-fold CV	81.92	—	—	—	—	—
Searle et al. [15] — DistilBERT + LR	ADReSS, $n = 48$	81.25 (68.1–89.8)	80.00	83.33	81.63	79.17	0.626
Luz et al. [7] — ADReSS linguistic baseline	ADReSS, $n = 48$	75.00 (61.2–85.1)	83.33	62.50	71.43	87.50	0.516

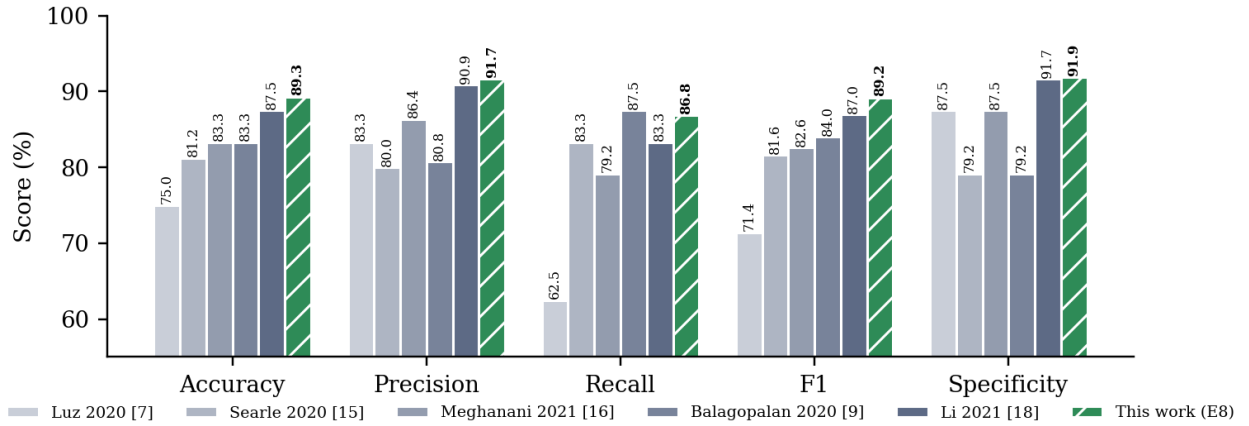


Figure 10. Metric-wise comparison of E8 with published systems that report per-class results. Syed et al. [12] and Fraser et al. [6] report accuracy only and are omitted.

The comparison also suggests where further gains are likely. Our transcripts keep lexical disfluencies but do not encode unfilled pause location or duration, which Yuan et al. [10, 11] showed to be a strong signal. Adding explicit pause tokens is therefore the highest-value data-side change available to us, and it requires no architectural work.

6.3 Theoretical Implications

Three implications seem worth drawing out, each weakly supported by a single experiment. First, the pooling results are consistent with aggregation acting as a genuine inductive bias in low-resource document classification: both attention pooling and hybrid pooling were associated with gains at negligible parameter cost, and combining selective and global summaries appears more useful than either alone. Second, the E5 result is consistent with the position of Mosbach et al. [31] that small-dataset fine-tuning behaviour is substantially an optimisation phenomenon, although the bundled design and unverified partition make this suggestive at best. Third, the class-weighting result is evidence that interventions without a principled justification for the data at hand deserve suspicion even when they improve a validation score.

6.4 Practical Implications

A system of this kind would be a triage aid, not a diagnostic instrument, and that framing decides which metric governs. In triage, a false negative means someone with AD is not flagged and not referred, while a false positive resolves at the cost of one further clinical assessment. E8 misses 5 of 38 AD documents (13.2%), against 8 or 9 for E5 (21.1–23.7%) and 14 for E1 (36.8%), while also producing the fewest false alarms (3 of 37 healthy speakers). That miss rate is still too high for autonomous use, and nothing in this study establishes that the system would perform at this level on unseen speakers, on a different corpus or on automatic transcripts. Threshold calibration deserves explicit attention: moving the operating point below $p = 0.5$ trades precision for recall deliberately.

For practitioners building similar pipelines, the procedural lessons are more transferable than any headline number: preserve disfluencies, consider learned and hybrid pooling before default pooling, examine the optimisation regime before reaching for architectural complexity, log data-table versions and test indices for every run, persist per-sample predictions so that paired tests remain available, and do not trust a validation optimum computed on 75 documents.

6.5 Unexpected Findings

Two results surprised us. The first is the scale of E3’s failure. The class weights sit very close to one, yet E3 produced the study’s best validation accuracy alongside a 2.67-point lower test accuracy than the unweighted attention model. At a validation size of 75, three or four documents falling favourably is worth about five points, so a 92.00% validation figure is fragile; its own Wilson interval runs from 83.6% to 96.3%. On near-balanced data there was no principled reason to apply class weighting in the first place.

The second is the direction of E8’s validation–test gap. E8’s best validation accuracy was 77.33%, lower than E2’s, yet it produced the highest test accuracy. Had model selection across configurations been based on validation scores, E8 would not have been chosen. This reinforces the point that a single 75-document validation set cannot reliably rank configurations whose true performance differs by a few points, and that the test figure itself carries comparable uncertainty.

6.6 A Reproducibility Flag

E5. The recorded test partition holds 37 HC and 38 AD documents (Table 2), but the E5 confusion matrix has row sums of 36 and 39. Two hypotheses are consistent with the data. H1: a single-document misassignment within the sealed partition, under which Table 5 applies and E5 accuracy lies between 81.33% and 84.00%. H2: a different split realisation, under which E5 is not comparable with E1 and E2 and must be re-run. We do not choose between them.

E8. The E8 confusion matrix sums correctly to 37 and 38, but the run used a 496-row table (240 HC) rather than the 497-row table (241 HC) of E1–E5. With a different number of input rows, a seeded stratified shuffle produces a different permutation, so the 75 test documents are unlikely to be identical to those used for E1 and E2 even though the class counts match. The E8 metrics are internally consistent and the test partition was consulted once, so the result is a valid held-out estimate on its own split; what is not established is that the E1-to-E8 difference is a within-split comparison.

Three checks resolve both questions. Re-derive the test indices from each table with `random_state = 42` and compare them document by document. Re-run E8 on the 497-row table (or E1 and E2 on the 496-row table) with test indices logged. And persist per-sample (index, true label, predicted label) triples for every configuration, which additionally enables the McNemar test that Section 5.8 could only bound.

7. LIMITATIONS

Unverified E5 and E8 partitions. E5 is not verified to have been scored on the sealed test partition, and E8 was scored on a count-matched partition drawn from a table one row smaller. Every E5 and E8 comparison in this paper carries that status.

Possible speaker-level leakage. This is the most serious risk to external validity. The Pitt corpus is longitudinal, and many participants contributed several Cookie Theft descriptions across annual visits [1]. If different sessions from the same speaker land in both training and test partitions, a model could earn part of its test accuracy by recognising speaker-specific style rather than disease-related language. The derived table carries no participant identifiers, so we could not verify disjointness and do not claim it.

Single seed, single split. Each configuration was trained once. Dodge et al. [32] report that initialisation and data ordering alone shift accuracy by several points on small datasets, and Yuan et al. [11] found run-to-run variance large enough to require ensembling. E8’s 12-point validation–test inversion is a concrete example of that variability.

Test-set size. One document is worth 1.33 points. The more configurations are compared against one fixed partition, the more it becomes a de facto validation set through accumulated selection decisions.

Bundled factors. E5 varies four optimisation factors simultaneously, and E8 adds mean pooling and raises dropout together. Neither row isolates a single factor.

No cross-validation, no probability outputs, no explainability analysis. No measure of split-to-split variability, no ROC, PR or calibration assessment, and no validated token attributions are available.

Manual transcripts, truncation and single corpus. The model consumes manual CHAT transcripts [2]; performance on ASR output would be lower [8]. A 256-token cap truncates the

longest descriptions. Everything reported here comes from one corpus collected at one site.

8. CONCLUSION AND FUTURE WORK

We ran a controlled five-configuration ablation of transcript-only Alzheimer’s disease classification with DeBERTa-v3 on Pitt DementiaBank Cookie Theft transcripts. Adding a learned attention-pooling head was associated with held-out accuracy rising from 74.67% to 78.67% and MCC from 0.510 to 0.588, a comparison verified at both ends. Concatenating attention-pooled and mean-pooled representations (E8) produced the best result of the study: 89.33% accuracy, 91.67% precision, 86.84% recall, 91.89% specificity, 89.19% F1 and MCC 0.788, with only eight errors in 75 documents and gains in both sensitivity and specificity. A bundled regularization regime recorded 82.67% (81.33–84.00% under sensitivity analysis), and class-weighted cross-entropy produced the highest validation accuracy but a lower held-out accuracy.

What these numbers support is narrower than the headline. The E8 run changes pooling and dropout together, was run once, and used a table one row smaller than the other configurations, so its test documents are not verified to match theirs. The E1-to-E8 difference would be statistically robust if the partitions match; the other comparisons are not separable at $n = 75$; and a possible speaker-leakage confound remains unexcluded. The defensible claim is that hybrid attention–mean pooling is associated with a substantial improvement in this setting and is the configuration most worth confirming.

8.1 Confirmation Protocol

- 1. Partition audit and E8 re-run.** Re-run E8 on the 497-row table with test indices logged, and re-run E5 under the same audit.
- 2. Speaker-disjoint evaluation.** Recover participant identifiers and repeat the ablation under grouped k-fold cross-validation with no speaker in more than one fold.
- 3. Multi-seed, cross-validated estimates.** Five seeds across ten folds per configuration, reporting mean and standard deviation [32, 42].
- 4. Paired significance testing.** Persist per-sample predictions and run McNemar’s exact test [37, 38] with Holm correction across the configuration family.
- 5. Factorial decomposition.** Separate mean pooling from dropout in E8, and learning rate, weight decay, schedule length and warmup in E5.
- 6. External validation.** Evaluate the frozen E8 configuration on the ADReSS test set [7] without retuning.

8.2 Extensions

- 1. Pause and disfluency encoding.** Insert explicit tokens marking unfilled-pause location and duration [10, 11]; on the published evidence, the largest expected gain available.

2. **Ensembling.** Average predictions across seeds and folds, which Yuan et al. [11] found necessary for stable results.
3. **Attention-weight interpretability.** Extract α for test documents, aggregate by predicted class, and cross-check against LIME or SHAP [44–47].
4. **Calibration and operating points.** Retain probabilities; report ECE, reliability diagrams and full ROC and PR curves [43]; select the threshold against an explicit cost ratio.
5. **Sequence length.** Ablate 512 tokens against 256 to price the truncation.
6. **Multimodal fusion.** Combine the transcript encoder with paralinguistic acoustic features [13, 18].

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Conflict of Interest. The authors declare that they have no competing financial or non-financial interests.

Ethics Statement. This work is a secondary analysis of an existing, de-identified, publicly documented corpus distributed under the DementiaBank access conditions of the TalkBank project. No new human-subject data were collected and no participant was re-identified. The original collection was conducted under the University of Pittsburgh Alzheimer Research Program protocol [1]; DementiaBank is supported by NIH-NIDCD grant R01-DC008524. The system described here is a research artefact. It is not a diagnostic device, has not been clinically validated, and must not be used to inform patient care.

Data Availability. The Pitt corpus is available to DementiaBank members through TalkBank at <https://talkbank.org/dementia/access/English/Pitt.html>. The derived 497- and 496-row tables, split indices, configuration files, per-sample predictions and model definition are available from the corresponding author on reasonable request.

Author Contributions. Following the CRediT taxonomy. Bhanu Prakash: [conceptualisation / methodology / software / investigation / writing – original draft]. Krishitha Myneni: [data curation / software / validation / formal analysis / writing – original draft]. Laasya Priya: [investigation / validation / visualisation / writing – review and editing]. All authors read and approved the final manuscript. [Assign the bracketed roles to the appropriate author before submission.]

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